

Foundation (Jalapeno)

- Q1.** The 4 in 425,317 represents 4×10^5 . What is the value of the 2 in this number?
(A) 2×10^0
(B) 2×10^2
(C) 2×10^4
(D) 2×10^6
(E) 2×10^7
- Q2.** Rounding 732.456 to the nearest 10 yields what result?
(A) 730
(B) 732
(C) 740
(D) 732.5
(E) 732.46
- Q3.** On Mars, 4,239 kilograms of space rocks were collected. What is the value of the 3 in this number?
(A) 3×10^2
(B) 3×10^1
(C) 3×10^0
(D) 3×10^3
(E) 3×10^4
- Q4.** 6,428 in expanded form is $6 \times 10^3 + 4 \times 10^2 + 2 \times 10^0 + 8 \times 10^1$. True or False?
(A) True
(B) False
(C) Maybe
(D) Not enough information
(E) None of the above
- Q5.** What is the standard form of $5 \times 10^2 + 2 \times 10^1 + 9 \times 10^0$?
(A) 529
(B) 592
(C) 523
(D) 509
(E) 502
- Q6.** A lab experiment involves 0.0425 grams of a substance. Rounded to the nearest 0.01 grams, what is the result?
(A) 0.04 grams
(B) 0.042 grams
(C) 0.043 grams
(D) 0.0425 grams
(E) 0.05 grams
- Q7.** 934.27 rounded to the nearest 10 is
(A) 930
(B) 940
(C) 934.3
(D) 934
(E) 935
- Q8.** In a space mission, 145.9 kilometers traveled per hour is recorded. Rounding this to the nearest 10 yields what result?
(A) 150 kilometers
(B) 140 kilometers
(C) 145.9 kilometers
(D) 145 kilometers
(E) 149 kilometers

Open Ended Questions (FRQ)

- Q9.** A scientist collects 3,456,219 space particles. Write this number in expanded form.
- Q10.** The distance between Earth and a planet is 4,239,027 kilometers. Round this to the nearest 100,000 kilometers and explain your reasoning.

Chef's Tips

- Emphasize the importance of place value in multi-digit numbers
- Have students practice converting between standard and expanded forms
- Remind students to read the questions carefully and choose the correct answer based on the given options